Example 10.4

A soft drink company has two bottling plants C_1 and C_2. Each plant produces three different soft drinks S_1,S_2 and S_3. The production of the two plants in number of bottles per day are:

Product	Plant	
	C_1	C_2
S_1	3000	1000
S_2	1000	1000
S_3	2000	6000

A market survey indicates that during the month of April there will be a demand for 24000 bottles of S_1, 16000 bottles of S_2 and 48000 bottles of S_3. The operating costs, per day, of running plants C_1 and C_2 are respectively ₹ 600 and ₹ 400. How many days should the firm run each plant in April so that the production cost is minimized while still meeting the market demand? Formulate the above as a linear programming model.

Solution:

(i) Variables:

Let x_1 be the number of days required to run plant C_1 and x_2 be the number of days required to run plant C_2.

Objective function: Minimize

$Z = 600\ x_1 + 400\ x_2$

(ii) Constraints: $3000\ x_1 + 1000\ x_2 \geq 24000$ (since there is a demand of 24000 bottles of drink A, production should not be less than 24000)

$1000\ x_1 + 1000\ x_2 \geq 16000$

$2000\ x_1 + 6000\ x_2 \geq 48000$

(iii) Non-negative restrictions: Since be the number of days required of a firm are non-negative, we have $x_1, x_2 \geq 0$.

Thus we have the following LP model.

Minimize $Z = 600\ x_1 + 400\ x_2$ subject to $3000\ x_1 + 1000\ x_2 \geq 24000$

$1000\ x_1 + 1000\ x_2 \geq 16000$

$2000\ x_1 + 6000\ x_2 \geq 48000$ and $x_1, x_2 \geq 0$

10.1.2 Solution of LPP by graphical method

After formulating the linear programming problem, our aim is to determine the values of decision variables to find the optimum (maximum or minimum) value of the objective function. Linear programming problems which involve only two variables can be solved by graphical method. If the problem has three or more variables, the graphical method is impractical.

The major steps involved in this method are as follows:

(i) State the problem mathematically

(ii) Write all the constraints in the form of equations and draw the graph

(iii) Find the feasible region

(iv) Find the coordinates of each vertex (corner points) of the feasible region. The coordinates of the vertex can be obtained either by inspection or by solving the two equations of the lines intersecting at the point

(v) By substituting these corner points in the objective function we can get the values of the objective function

(vi) If the problem is maximization then the maximum of the above values is the optimum value. If the problem is minimization then the minimum of the above values is the optimum value.

10_11th_BM-STAT_Ch-10-EM.indd 240 21-04-2020 12:32:49 PM